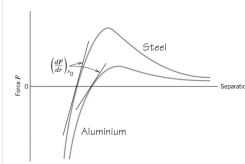
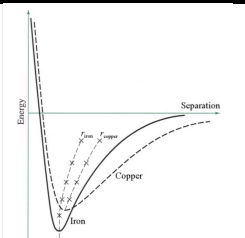


Structure Property Relationships			Atomic Properties		
Structure	Property	Reason/Further Properties			initial slope shows force needed to separate atoms thus slope is indicative of the young's modulus young's modulus governs the stiffness/deflection resistance thus higher slope causes higher stiffness
Metals			Stiffness (force-separation curve)		
closely packed crystalline structure	High density	<i>due to</i> heavy metal atoms			
	High thermal & electrical conductivity				
sea of delocalised electrons	High malleability and ductility	<i>due to</i> layers that can slide past one another			
	Low corrosion resistance	<i>due to</i> easily removable electrons			
Ceramics			Thermal Expansion (energy-separation curve)		heating causes ↑KE, thus ↑U potential shifts up energy well well gets wider, thus more vibration about mean pos well gets asymmetric, mean position is further apart expands as temp increases
	high corrosion resistance	<i>due to</i> hard to overcome strong ionic/covalent bonds			
ionic/covalent bonded crystalline structure	↳ low shear to fracture				thinner, deeper wells require more heating and energy to move up well to vibrate the same amnt thus lower thermal expansion
	↳ low plastic deformation	contributes to brittle, low toughness, high hardness			
Amorphous Thermoplastics			Metal Strengthening (prevent motion of dislocations)	Plastic Deformation from dislocations	shear force applied dislocation moves across a plane along the crystal structure
covalently bonded mers free to rotate	low stress to deform/low yield stress	<i>due to</i> the coils tangling, which then require low stress to slide apart			
adjacent strands bonded by VDW or hydrogen bonds	(for H bonds) higher yield strength, stiffness and toughness			1. Grain Size Reduction	smaller grain size --> more grain boundaries boundaries restrict movement of dislocations more stress needed to overcome (↑yield strength)
<i>may have</i> larger side groups	higher yield strength, stiffness, lower ductility	<i>due to</i> the coils being harder to untangle		2. Solid Solution Strengthening/ Precipitation Hardening	alloying elements added dissolves into alloy(SSS)/forms precipitates with alloy (PH) distorts regular alloy structure prevents movement of dislocations, more stress needed
	higher yield strength, stiffness, lower ductility when worked below glass transition temp	due to insufficient energy to have strands uncoil and slide past one another		3. Work Hardening/Cold Working	plastic deformation causes more dislocations to form further deformation causes dislocations to tangle more stress needed to overcome each other
Semi-Crystalline (regions of) Thermoplastics			Ceramic/Composite Toughening (increase energy absorbed)	1. Transformation Toughening	under stress, ceramic structure changes ↳ process absorbs energy volume increases, clamps cracks ↳ more energy needed to propagate cracks
crystalline structure with VDW or hydrogen bonds	higher yield strength, stiffness, lower ductility	<i>due to</i> the crystalline regions require more stress to align them in the stressed direction and to overcome the secondary bonds		2. Crack Deflection	fibers increase distance for crack to propagate ↳ dissipates energy more energy needed to fracture
Thermosets & Elastomers					
covalent cross links between chains	higher yield strength, stiffness, lower ductility	<i>due to</i> chains being unable to slide past one another		3. Crack Bridging & Fiber Pullout	crack propagates until meets fiber-matrix interface fibers transfer stress, reduces crack propagation energy dissipated into fiber pull out and debonding for further fracture more energy needed to propagate past fibers (slower growth)